

Can raw AIFS rainfall forecasts anticipate electricity-supply disruption during Indian floods?

Methods, results and limitations of a first complete analysis (ESMI 2014–2018)

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1 Summary

Question. How far ahead can raw ECMWF AIFS precipitation forecasts (lead times of 1–5 days) identify days of elevated electricity outages at Prayas ESMI voltage monitors during Indian flood events? And how much skill is lost relative to observed IMD rainfall, the “perfect-rainfall” benchmark?

Main findings.

1. **Rain and outages are linked, but only weakly.** Heavier rain significantly raises the odds of a disruption day: odds ratio ≈ 1.3 per e -fold increase in $(1 + \text{rain})$, $p < 10^{-4}$, with the same result in a model with a random intercept per station (Table 4). In genuinely out-of-sample tests (leave one flood episode out), the skill is small: ROC-AUC ≈ 0.56 – 0.57 on flood-event windows, against 0.54 for climatology, and Brier skill score (BSS) ≈ 0.005 – 0.01 (Table 5).
2. **Skill hardly declines from day 1 to day 5.** AIFS rainfall itself degrades with lead: at the 3×3 level its correlation with IMD falls from 0.73 to 0.66 (Table 3). But the rain→outage signal is so weak that outage skill stays flat (AUC 0.573 at day 1, 0.572 at day 5).
3. **Replacing observed rain with the forecast costs almost nothing during floods.** On event windows, AIFS at every lead matches IMD within noise. Over all June–September (JJAS) days, IMD is modestly better (AUC 0.529 against 0.514 – 0.511).
4. **Persistence wins at day 1; AIFS wins at days 2–5.** Yesterday’s outage is the best single predictor at lead 1, and adding AIFS rain improves it (AUC $0.591 \rightarrow 0.615$). From lead 2 onward, the latest known outage (day $D - L$) carries no information, but AIFS keeps its small skill.
5. **Urban, peri-urban and rural stations do not differ beyond noise.** Rural stations look best on flood-event windows, but the rural and peri-urban event samples are small and the intervals overlap (Table 7).

The confidence intervals are wide (only 9 independent flood episodes), and several BSS intervals include zero. The results describe a *weak, marginally significant* signal, not usable operational skill.

2 Study design

The design follows `docs/TASK.md`:

- **Units of analysis.** Station-days at ESMI monitors, 2014–2018.
- **Flood events.** Global Flood Database (GFD; Tellman et al. 2021) events that inundate land near a station during its ESMI coverage. This rebuilds the event set of Verschuur & Balakrishnan (2024), who did not publish their list.
- **Predictors.** Daily rainfall at each station from AIFS at leads 1–5 days and, as a benchmark, from IMD gauge analyses.
- **Outcome.** A binary “disruption day” (outage minutes above the station’s 90th percentile).
- **Models.** Logistic models with climatology, persistence and IMD baselines.
- **Validation.** Leave-one-flood-episode-out cross-validation, so every score describes a flood the model has never seen, plus a leave-one-year-out analysis over the whole monsoon.

3 Data

Table 1: Data sources.

Dataset	Content	Notes
Prayas ESMI voltage	Minute-wise voltage, 10 CSVs, 2014–2018, 529 locations (meta-data)	Wide format (one row per location-hour). Five files were truncated downloads and were re-downloaded from Harvard Dataverse (doi:10.7910/DVN/CLLZZM); all now match Dataverse byte for byte.
ESMI location information	District, state, category, connection type, From/To dates	No coordinates; categories map to urban / peri-urban / rural.
IMD gridded rainfall	0.25° daily gauge analysis (Pai et al. 2014), 2014–2018	24 h totals ending 08:30 IST (03 UTC) on the labelled date.
ECMWF AIFS forecasts	1,818 daily 00 UTC initialisations (8 missing), 6-hourly steps to 120 h, global 0.25°	Read-only source (8.5 TB). India precipitation subset: 2.8 GB.
Global Flood Database	MODIS 250m flood footprints; 26 events in and around India, 2014–2018	Bands: <code>flooded</code> , <code>duration</code> , <code>clear_views</code> , <code>clear_perc</code> , <code>jrc_perm_water</code> .

4 Methods

4.1 ESMI processing (`scripts/01_esmi_profile_daily.py`)

The minute data were reshaped into daily, per-location totals:

- **Outage minute:** voltage < 130 V, following Verschuur & Balakrishnan. Exact 0 V could also be a monitor fault, but the two cannot be told apart, so 0 V counts as an outage (owner decision).
- **Observed day:** $\geq 90\%$ of the 1,440 minutes recorded. Missing rows are treated as monitor downtime, not as outages. On other days the outage metrics are set to missing.
- **Stations kept:** Agriculture connections are dropped, because farm feeders are rostered (owner decision). Stations labelled `[Offline]` are kept, using only data within their From–To dates.

The result is 517 stations and 311,778 station-days inside coverage, of which 85.3% are observed.

4.2 Station coordinates (`scripts/02_geocode_locations.py`)

ESMI publishes only locality names with district and state. Locations were geocoded with Nominatim, with a GeoNames fallback, constrained to the district. Of 529 stations:

- 300 are placed at the locality itself;
- 212 fall back to the city centroid;
- 17 fall back to the district centroid.

Each station is mapped to its nearest IMD cell with data. For 11 coastal stations the nearest cell has no data, so the nearest cell with data is used instead (Figure 1).

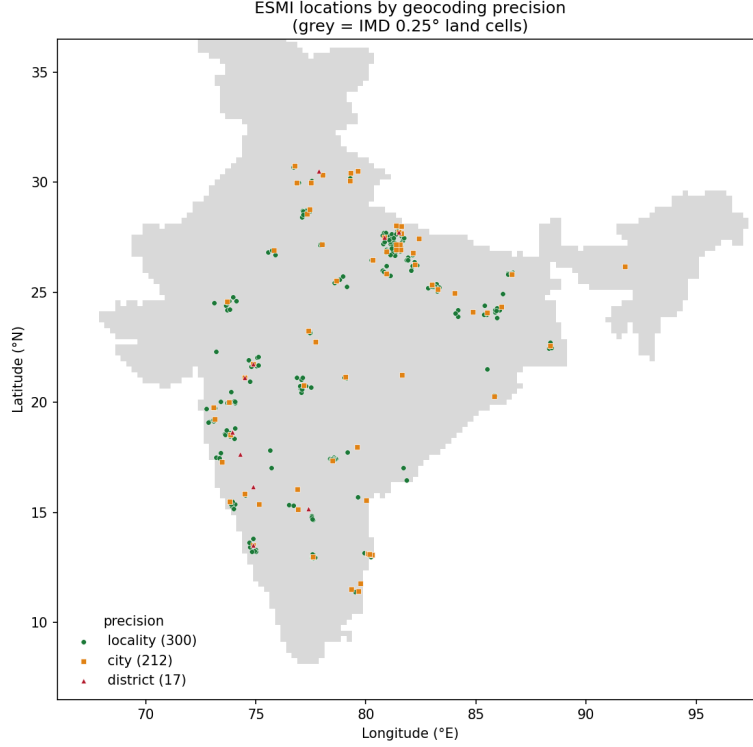


Figure 1: Geocoded ESMI stations (`scripts/02`; review list in `outputs/02_geocode_review.csv`).

4.3 AIFS pre-processing and checks (`scripts/03, 05`)

Only `total_precipitation` over 5–40°N, 65–100°E was extracted (141×141 points) from every initialisation file, with NCO. The source files were never modified. Three checks were made:

- **Units: metres.** Over Indian land in July 2016, the median ratio of IMD daily rain (mm) to the AIFS lead-1 daily sum is 896. A ratio of order 1,000 means metres; a ratio below 1,000 means AIFS is about 12% wetter than IMD over land ($1000/896 \approx 1.12$), consistent with the station-level wet bias in Section 5.1. The mean spatial correlation for day 1 is 0.561.
- **Step accumulation.** Values do not grow with lead time, so each step is a 6-hourly accumulation, not a running total from initialisation.
- **Initialisation and grid.** `time = 0` days since the init date and `leadtime = 6, 12, ..., 120` h, so initialisation is at 00 UTC. The IMD grid points (6.5, 6.75, ...°N; 66.5, ...°E) coincide exactly with AIFS points, so no regridding was needed.

4.4 Time alignment (`scripts/06_build_panel.py`)

IMD and ESMI use Indian Standard Time (UTC+5:30); AIFS uses UTC.

- **IMD day D** runs from 03 UTC on $D - 1$ to 03 UTC on D .
- **AIFS lead L for IMD day D** is the forecast initialised at 00 UTC on $D - L$, covering hours $24(L - 1) + 3$ to $24L + 3$ after initialisation. The two 6-hour steps straddling 03 UTC are split in half, assuming uniform rain within a step:

$$\text{AIFS}_L(D) = \frac{1}{2}s_{24L-18} + s_{24L-12} + s_{24L-6} + s_{24L} + \frac{1}{2}s_{24L+6},$$

where s_h is the step ending h hours after initialisation.

- **Lead 5 is approximate.** The forecast stops at 120 h, so the step ending at 126 h does not exist. The 21 available hours are scaled by 24/21.
- **48-hour totals** add days $D - 1$ and D from the same forecast run, so they exist for leads 2–5 only.
- **The ESMI day** (a calendar day in IST) runs from 18:30 UTC on $D - 1$ to 18:30 UTC on D . It overlaps the IMD day by 15.5 h, and the IMD day leads it by 8.5 h. IMD lags 0–2 are included to allow for this offset.

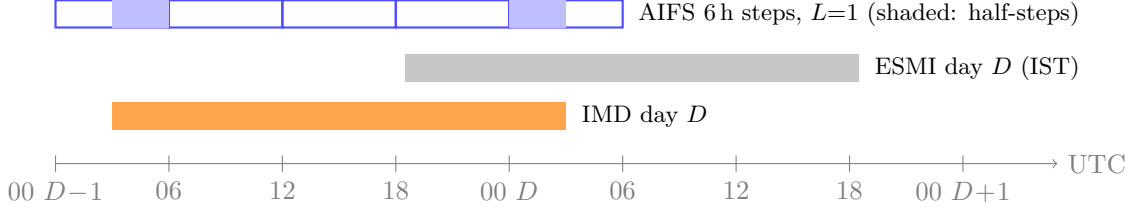


Figure 2: Time windows. For lead $L > 1$ the AIFS steps are those of the forecast initialised L days earlier.

4.5 Spatial matching

The centre cell is the nearest IMD cell with data; the same cell is used for AIFS. Three rain predictors were computed: the centre cell, the 3×3 mean ($\pm 0.25^\circ$, about 25 km) and the 3×3 maximum. Both 3×3 versions use IMD-valid cells only, for both sources. The 3×3 mean was fixed as the headline predictor *before* any results were seen; the other two are sensitivity checks.

4.6 Flood-event reconstruction (`scripts/04_match_flood_events.py`)

Verschuur & Balakrishnan report 15 GFD events overlapping ESMI stations but did not publish the list, and the authors declined to share it (they described the data quality as poor). The events were therefore rebuilt from the 26 GFD footprints in and around India (Table 2):

1. **Flooded pixel:** GFD `flooded` = 1 and not JRC permanent water.
2. **River-channel mask.** A first pass showed that `flooded` includes seasonally wet river channels that JRC does not count as permanent. For example, the 11 Guwahati stations, 1–2 km from the Brahmaputra, “matched” 6 unrelated events, including one in March. Every footprint was therefore resampled (nearest neighbour) onto a common 250 m grid around each station, and pixels flooded in ≥ 3 of the 26 events were masked as recurrent river water (owner decision).
3. **Affected station:** $\geq 1 \text{ km}^2$ of masked flooding within 20 km. The paper’s plain rule (any flooded pixel within 20 km) is kept as a sensitivity column.
4. **Event window:** GFD start -3 days to GFD end $+3$ days.
5. **Completeness:** the whole window must lie inside the station’s ESMI coverage, and every day must be observed. This implements the rule “skip the event if observations are missing”.
6. **Episodes:** events whose windows overlap in time are merged into one episode for cross-validation (4378+4382 in 2016; 4507+4508 in 2017), because they share the same rain and the same station-days.

The result is 11 events in 9 episodes, with 124 station-events at 84 stations. The plain rule would give 12 events and 226 station-events. Of the kept stations, 40 are placed at a city centroid, not the locality itself.

Table 2: All 26 GFD footprints considered. “Affected”: stations with $\geq 1 \text{ km}^2$ of masked flooding within 20 km. “In cov.”: of those, stations whose ESMI coverage overlaps the event. “Kept”: the window is fully covered and completely observed. “Kept (raw)”: the same, under the plain 20 km rule.

ID	Country	Began	Ended	Cause	Affected	In cov.	Kept	Kept (raw)	Used
4215	Sri Lanka	2014-12-20	2015-01-01	Monsoonal Rain	0	0	0	0	
4239	India	2015-03-20	2015-03-31	Heavy Rain	0	0	0	0	
4251	India	2015-05-17	2015-05-21	Torrential Rain	28	2	1	1	yes
4259	India	2015-06-02	2015-06-29	Monsoonal Rain	11	0	0	0	
4265	India	2015-06-24	2015-06-29	Monsoonal Rain	0	0	0	0	
4272	Pakistan	2015-07-15	2015-08-19	Monsoonal Rain	0	0	0	0	
4282	India	2015-07-15	2015-08-19	Monsoonal Rain and Tropical Storm	98	5	2	7	yes
4283	Myanmar	2015-07-15	2015-08-19	Monsoonal Rain	0	0	0	0	
4288	India	2015-08-13	2015-09-11	Monsoonal Rain	11	0	0	0	
4309	India	2015-11-10	2015-12-04	Tropical Storm Roan	44	15	5	8	yes
4339	Pakistan	2016-03-12	2016-04-13	Heavy Rain	0	0	0	0	
4355	India	2016-04-20	2016-05-01	Heavy Rain	0	0	0	0	
4365	Myanmar	2016-06-01	2016-08-16	Monsoonal Rain	0	0	0	0	
4378	India	2016-07-07	2016-08-03	Monsoonal Rain	61	18	8	27	yes
4382	Bangladesh	2016-07-25	2016-08-26	Monsoonal Rain	116	32	18	27	yes
4459	Bangladesh	2017-03-30	2017-04-18	Heavy Rain	11	11	9	9	yes
4483	India	2017-06-02	2017-07-03	Monsoonal Rain	11	11	10	10	yes
4499	India	2017-07-27	2017-08-10	Monsoonal Rain	1	0	0	2	
4507	India	2017-08-10	2017-08-26	Monsoonal Rain	35	11	5	32	yes
4508	Bangladesh	2017-08-10	2017-08-26	Monsoonal Rain	46	22	13	40	yes
4632	Myanmar	2018-06-15	2018-06-20	Monsoonal Rain	0	0	0	0	
4640	India	2018-06-25	2018-07-11	Monsoonal Rain	0	0	0	0	
4645	Pakistan	2018-07-03	2018-07-11	Monsoonal Rain	0	0	0	0	
4666	Myanmar	2018-07-15	2018-08-10	Monsoonal Rain	0	0	0	0	
4665	India	2018-08-02	2018-08-10	Monsoonal Rain	11	6	5	5	yes
4673	India	2018-09-01	2018-09-07	Monsoonal Rain	159	79	48	58	yes

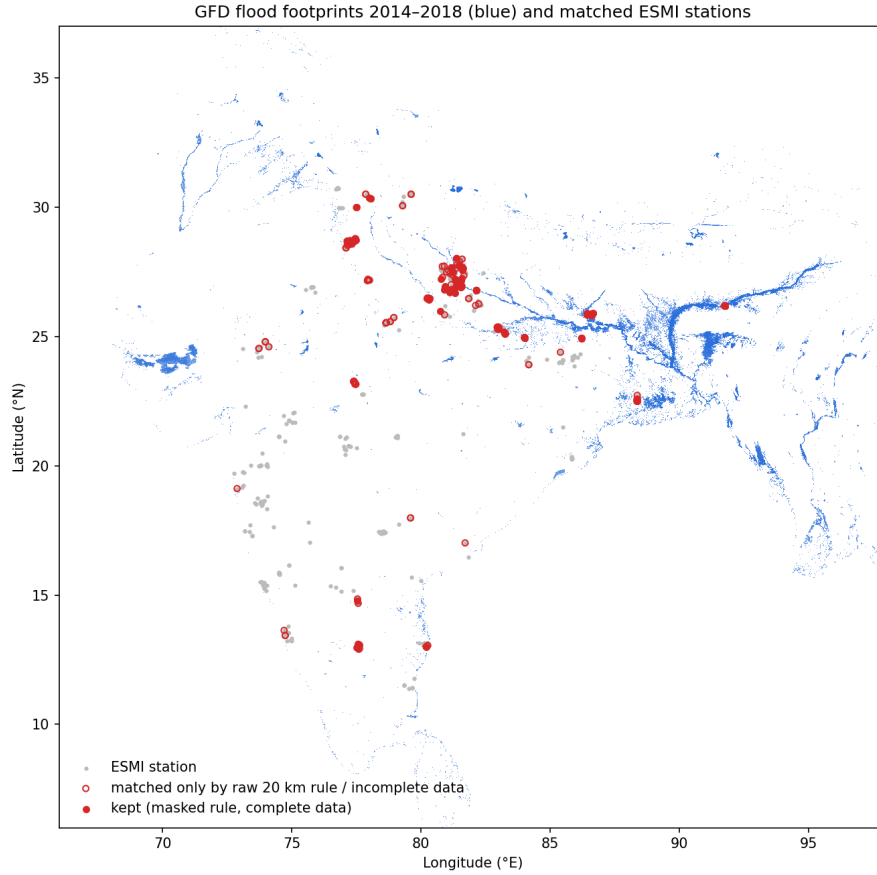


Figure 3: GFD footprints, 2014–2018 (blue), and matched stations. Filled red: kept under the masked rule with complete data. Open red: matched only by the plain rule, or with incomplete data.

4.7 Outcome definition

- **Disruption day:** daily outage minutes above the station’s 90th percentile. The percentile is computed over observed days that are not near any flood (plain-rule windows excluded).
- **Base rate:** 9.8% of all station-days, 12.3% in event windows and in JJAS. Floods and the monsoon bring more outages, and 13 stations whose 90th percentile is 0 count any outage as a disruption.
- **Secondary outcome:** excess minutes over the station’s same-month, same-weekday median. This is stored in the panel but not modelled in this pass.

4.8 Models and baselines (`scripts/07_model_skill.py`)

Every model is a binomial GLM with climatology entered as an offset:

$$\text{logit Pr}(\text{disrupt}_{s,D}) = \text{logit } p_{s,m(D)}^{\text{clim}} + \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_{s,D}.$$

Here p^{clim} is the station $\times$ month disruption rate from the training fold, shrunk toward the station rate (pseudo-count 20). Station, month and season are therefore controlled through the offset. Predictors $\mathbf{x}$:

- **Climatology:** none (the offset alone).
- **Persistence at lead L :** disruption on day $D - L$, the latest outage known when a lead- L forecast is issued.
- **AIFS 24 h:** $\log(1 + \text{rain})$ at lead L . **AIFS 48 h:** the same for days $D - 1$ and D from one forecast run ($L \geq 2$).
- **AIFS + persistence** at lead L .
- **IMD benchmarks:** same day, 48 h, and lags 0–2 (days D , $D - 1$, $D - 2$).

A random-intercept (per station) Bayesian mixed GLM, with month fixed effects fitted by variational Bayes, serves as an in-sample check. IFS forecasts for 2014–2018 are not available, so they are not included.

4.9 Cross-validation and scoring

- **Event analysis: leave one flood episode out (9 folds).** The test set is the held-out episode’s event-window station-days. *All dates* of that episode are removed from training for *every* station, to prevent leakage through spatially correlated days. Two training sets are used:
 - *event-only* (headline): the other episodes’ event windows;
 - *all days*: every other observed station-day.
- **Monsoon analysis: leave one year out** over JJAS 2015–2018, trained on all days.
- **Scores:** Brier score, BSS against climatology, ROC-AUC and reliability. All models are scored on the same rows (every predictor available).
- **Uncertainty:** 95% intervals from a block bootstrap, resampling episodes for the event analysis (1,000 draws) and stations for JJAS (200 draws).

5 Results

5.1 AIFS rainfall against IMD

Over JJAS station-days, AIFS is about 0.8–1.0 mm/day wetter than IMD at the centre cell and for the 3×3 mean. Its 3×3 maximum is about 6.5 mm/day *lower* than IMD’s, because the gauge-based

IMD field has sharper local peaks. The correlation with IMD falls steadily with lead (Table 3, Figure 4). So the forecast degrades as expected: it is the outage link that is weak.

Table 3: AIFS against IMD daily rain at the stations, JJAS.

Lead (d)	Pearson r with IMD			Bias AIFS–IMD (mm/day)		
	centre	3×3 mean	3×3 max	centre	3×3 mean	3×3 max
1	0.635	0.732	0.693	+0.78	+0.75	-6.32
2	0.626	0.721	0.682	+1.01	+0.99	-6.26
3	0.613	0.707	0.672	+0.97	+0.93	-6.47
4	0.596	0.688	0.655	+0.83	+0.79	-6.67
5	0.568	0.656	0.629	+0.91	+0.87	-6.62

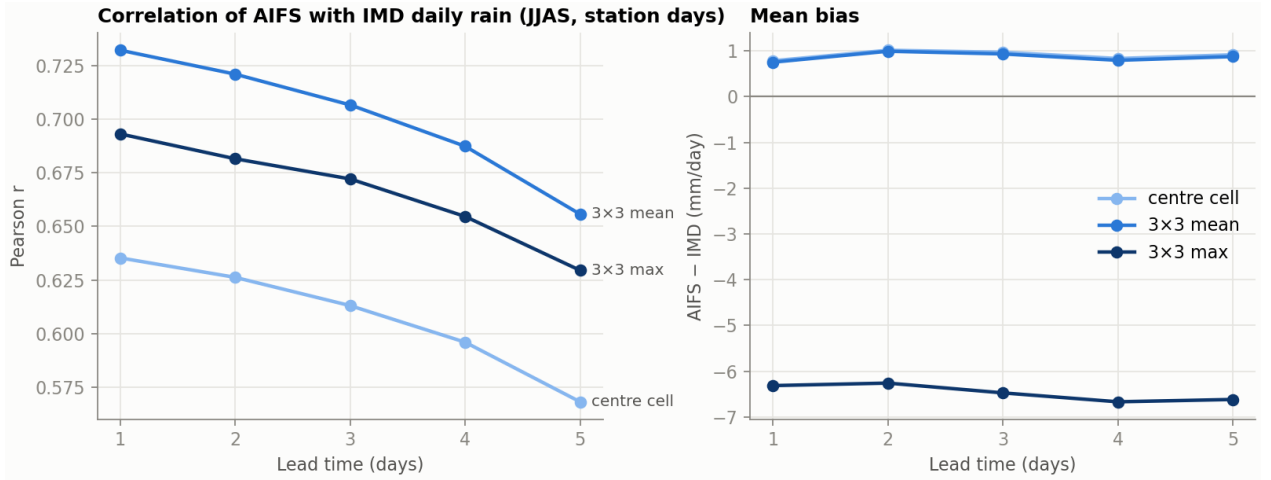


Figure 4: Correlation (left) and mean bias (right) of AIFS against IMD station rain, by lead and spatial aggregation.

5.2 Event time series

Figure 5 shows each episode. AIFS tracks the observed rain peaks at every lead, often with a wet bias (episodes 7–9). Outage minutes, however, are dominated by station-specific baseline levels and noise. For example, episode 4 begins at about 800 min/day, before the flood peak. The two Kolkata stations in episode 2 record no outages at all (CESC supply is very reliable).

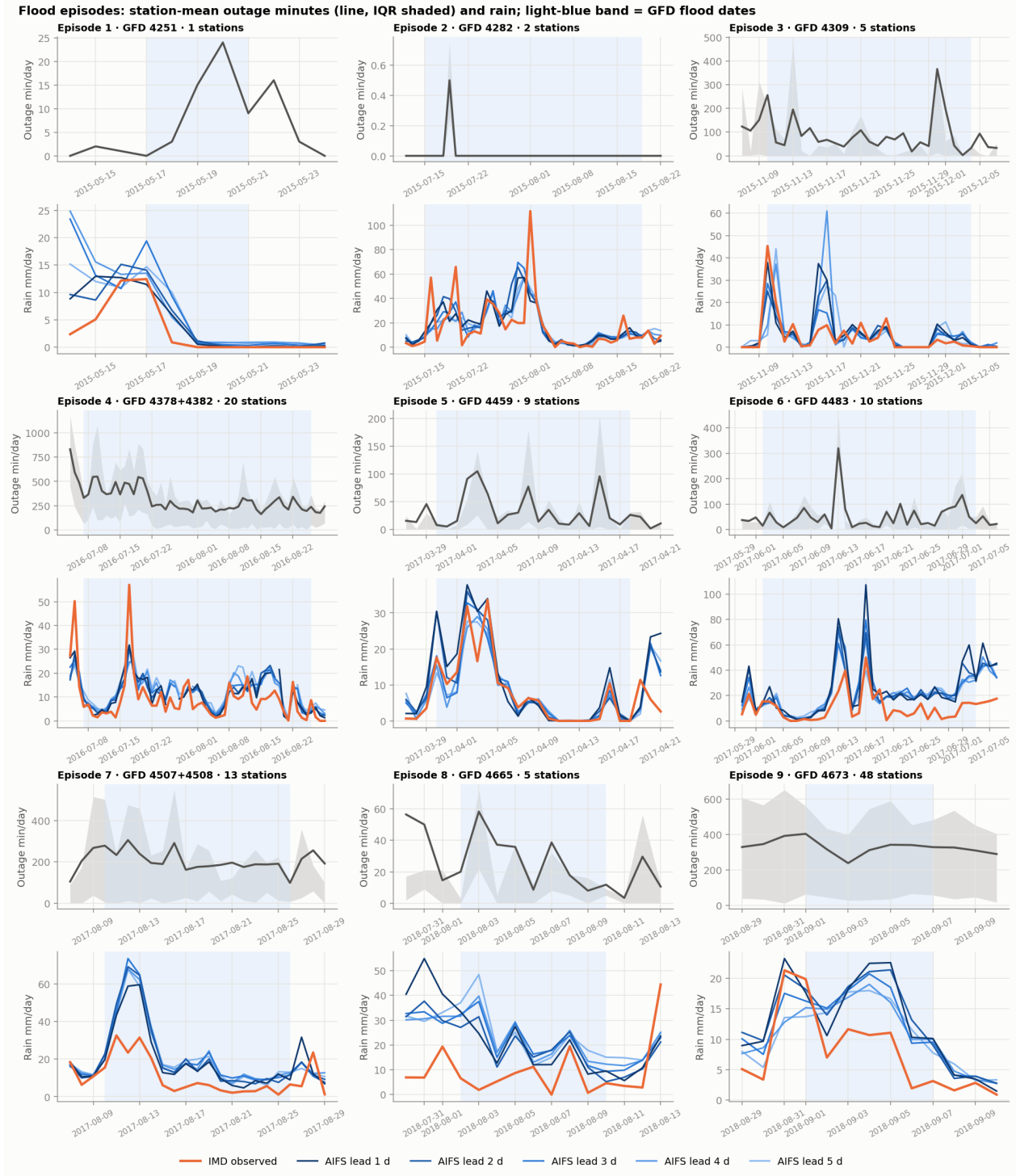


Figure 5: Per flood episode: station-mean outage minutes (line; interquartile range shaded) and station-mean rain from IMD (orange) and AIFS at leads 1–5 (dark to light blue). Light-blue band: GFD flood dates.

5.3 Rain–outage association

Table 4: In-sample coefficient on $\log(1 + \text{rain})$ over event-window station-days ($n = 2,508$). OR: odds ratio per unit of $\log(1 + \text{rain})$.

Predictor	GLM, climatology offset				Random intercept per station		
	β	SE	p	OR	β	SD	station SD
IMD, day D	0.239	0.056	<0.001	1.27	0.215	0.028	0.72
AIFS 24 h, lead 1	0.299	0.067	<0.001	1.35	0.301	0.023	0.71
AIFS 24 h, lead 2	0.308	0.072	<0.001	1.36	0.307	0.023	0.71
AIFS 24 h, lead 3	0.315	0.075	<0.001	1.37	0.308	0.023	0.71
AIFS 24 h, lead 4	0.313	0.077	<0.001	1.37	0.299	0.023	0.71
AIFS 24 h, lead 5	0.343	0.079	<0.001	1.41	0.331	0.023	0.72

The association is highly significant and consistent between the fixed and mixed models. The AIFS coefficients are slightly larger than IMD’s, so AIFS rain is at least as informative about outages as the gauge analysis during these events. The station random-intercept SD (≈ 0.7 on the logit scale) shows large differences between stations, which the climatology offset already absorbs.

5.4 Cross-validated skill

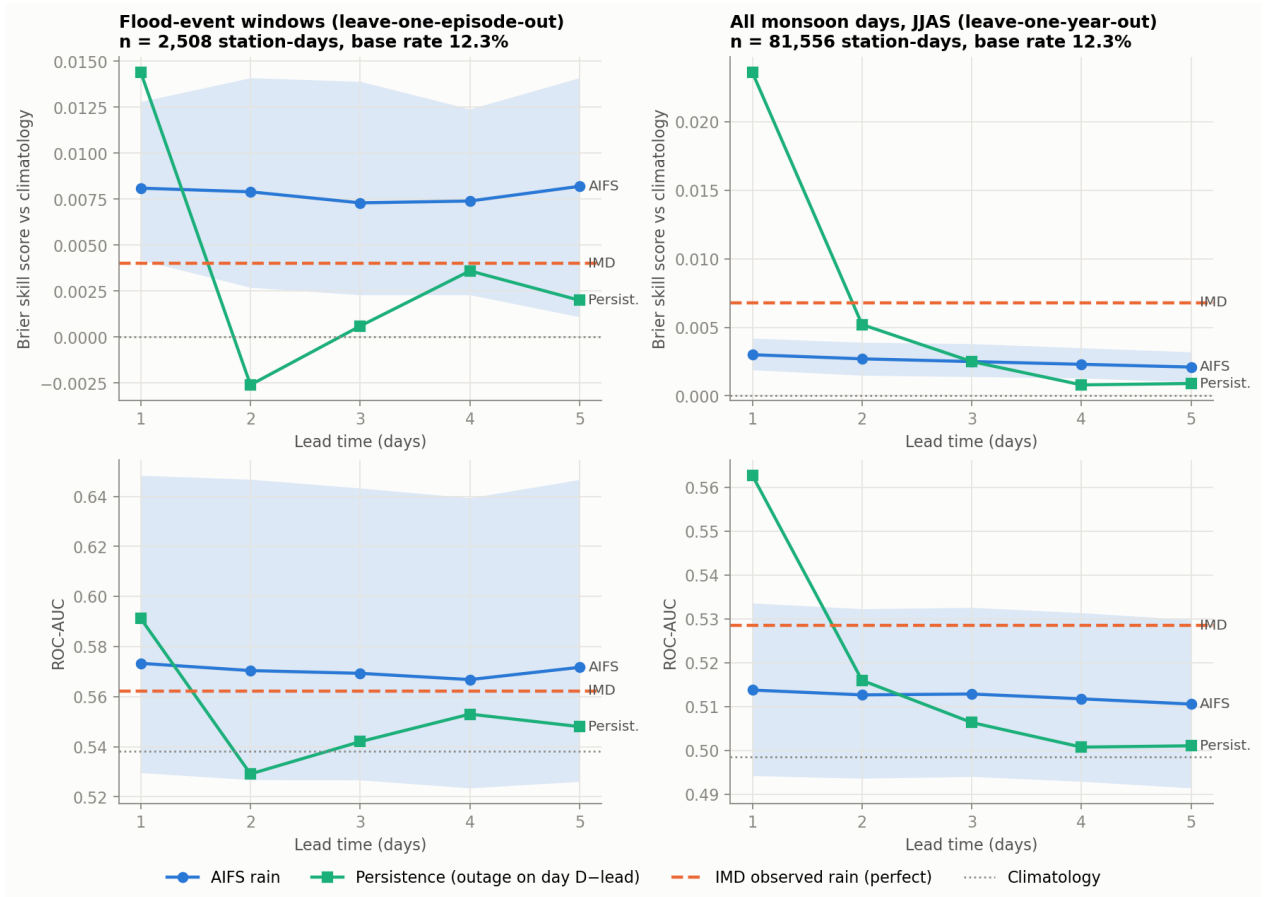


Figure 6: BSS (top) and ROC-AUC (bottom) against lead. Left: flood-event windows (leave one episode out, event-only training). Right: all JJAS days (leave one year out). The blue band is the 95% bootstrap interval for AIFS.

Table 5: Skill on flood-event windows, leave one episode out, event-only training ($n = 2,508$, base rate 12.3%).

Model	Brier	BSS	95% CI	AUC	95% CI
Climatology	0.1090	0.000	[0.000, 0.000]	0.538	[0.491, 0.612]
IMD, day D	0.1085	0.004	[-0.006, 0.011]	0.562	[0.515, 0.630]
IMD, 48 h	0.1089	0.001	[-0.008, 0.007]	0.549	[0.500, 0.623]
IMD, lags 0–2	0.1088	0.002	[-0.011, 0.009]	0.556	[0.510, 0.621]
Persistence, lead 1	0.1074	0.014	[0.002, 0.024]	0.591	[0.539, 0.656]
Persistence, lead 2	0.1092	-0.003	[-0.011, 0.001]	0.529	[0.482, 0.604]
Persistence, lead 3	0.1089	0.001	[-0.013, 0.007]	0.542	[0.491, 0.609]
Persistence, lead 5	0.1087	0.002	[-0.007, 0.007]	0.548	[0.502, 0.619]
AIFS 24 h, lead 1	0.1081	0.008	[0.004, 0.013]	0.573	[0.530, 0.648]
AIFS 24 h, lead 2	0.1081	0.008	[0.003, 0.014]	0.570	[0.527, 0.647]
AIFS 24 h, lead 3	0.1082	0.007	[0.002, 0.014]	0.569	[0.527, 0.643]
AIFS 24 h, lead 4	0.1082	0.007	[0.002, 0.012]	0.567	[0.523, 0.639]
AIFS 24 h, lead 5	0.1081	0.008	[0.001, 0.014]	0.572	[0.526, 0.647]
AIFS 48 h, lead 2	0.1086	0.003	[-0.001, 0.007]	0.555	[0.509, 0.632]
AIFS 48 h, lead 5	0.1085	0.004	[0.000, 0.008]	0.559	[0.513, 0.633]
AIFS + persistence, lead 1	0.1068	0.020	[0.006, 0.035]	0.615	[0.566, 0.683]
AIFS + persistence, lead 3	0.1082	0.007	[0.002, 0.014]	0.573	[0.529, 0.643]
AIFS + persistence, lead 5	0.1079	0.009	[0.004, 0.016]	0.582	[0.537, 0.653]

Table 6: Skill over all JJAS days, leave one year out ($n = 81,556$, base rate 12.3%).

Model	Brier	BSS	95% CI	AUC	95% CI
Climatology	0.1108	0.000	[0.000, 0.000]	0.498	[0.480, 0.519]
IMD, day D	0.1100	0.007	[0.005, 0.008]	0.528	[0.510, 0.547]
IMD, 48 h	0.1103	0.004	[0.003, 0.005]	0.518	[0.499, 0.537]
IMD, lags 0–2	0.1098	0.009	[0.007, 0.010]	0.536	[0.519, 0.553]
Persistence, lead 1	0.1082	0.024	[0.021, 0.027]	0.563	[0.547, 0.579]
Persistence, lead 2	0.1102	0.005	[0.004, 0.006]	0.516	[0.499, 0.535]
Persistence, lead 3	0.1105	0.003	[0.002, 0.003]	0.506	[0.489, 0.527]
Persistence, lead 5	0.1107	0.001	[0.001, 0.001]	0.501	[0.484, 0.522]
AIFS 24 h, lead 1	0.1104	0.003	[0.002, 0.004]	0.514	[0.494, 0.534]
AIFS 24 h, lead 2	0.1105	0.003	[0.002, 0.004]	0.513	[0.494, 0.532]
AIFS 24 h, lead 3	0.1105	0.003	[0.001, 0.004]	0.513	[0.494, 0.533]
AIFS 24 h, lead 4	0.1105	0.002	[0.001, 0.004]	0.512	[0.493, 0.531]
AIFS 24 h, lead 5	0.1105	0.002	[0.001, 0.003]	0.511	[0.491, 0.530]
AIFS 48 h, lead 2	0.1106	0.002	[0.001, 0.002]	0.507	[0.488, 0.526]
AIFS 48 h, lead 5	0.1106	0.001	[0.000, 0.002]	0.506	[0.487, 0.525]
AIFS + persistence, lead 1	0.1080	0.025	[0.022, 0.029]	0.572	[0.555, 0.587]
AIFS + persistence, lead 3	0.1102	0.005	[0.004, 0.006]	0.520	[0.503, 0.540]
AIFS + persistence, lead 5	0.1104	0.003	[0.002, 0.004]	0.513	[0.495, 0.532]

Reading the tables.

- On event windows every rain model beats climatology, but only by 0.005–0.01 in BSS. AIFS is flat across leads and on a par with IMD.
- Over all JJAS days, climatology has $AUC \approx 0.5$ by construction, because the outcome is station-relative, so every station has a similar base rate. There IMD beats AIFS, and AIFS declines slightly with lead.

- Persistence at lead 1 is the best single predictor in both samples, and adding AIFS rain improves on it. At leads ≥ 2 persistence falls to the level of climatology.
- The 48 h totals and lags 0–2 add nothing over the single day.

5.5 Reliability

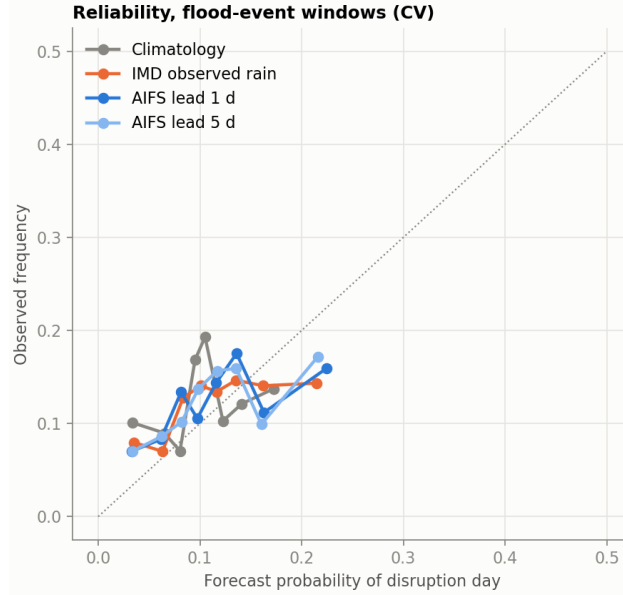


Figure 7: Reliability of the cross-validated event-window probabilities. The forecasts stay within ≈ 0.03 – 0.22 , so they are well calibrated but have little sharpness.

5.6 Urban, peri-urban and rural

Table 7: ROC-AUC by ESMI area type (3×3 mean).

Sample	Area	n	Clim.	IMD	Pers. L1	AIFS L1	AIFS L3	AIFS L5
Event windows	urban	1,833	0.550	0.552	0.599	0.572	0.570	0.571
Event windows	peri-urban	104	0.540	0.594	0.530	0.561	0.531	0.528
Event windows	rural	571	0.494	0.590	0.572	0.587	0.579	0.588
JJAS all days	urban	48,740	0.500	0.521	0.555	0.509	0.508	0.506
JJAS all days	peri-urban	10,990	0.507	0.544	0.578	0.528	0.527	0.525
JJAS all days	rural	21,826	0.488	0.537	0.571	0.517	0.516	0.512

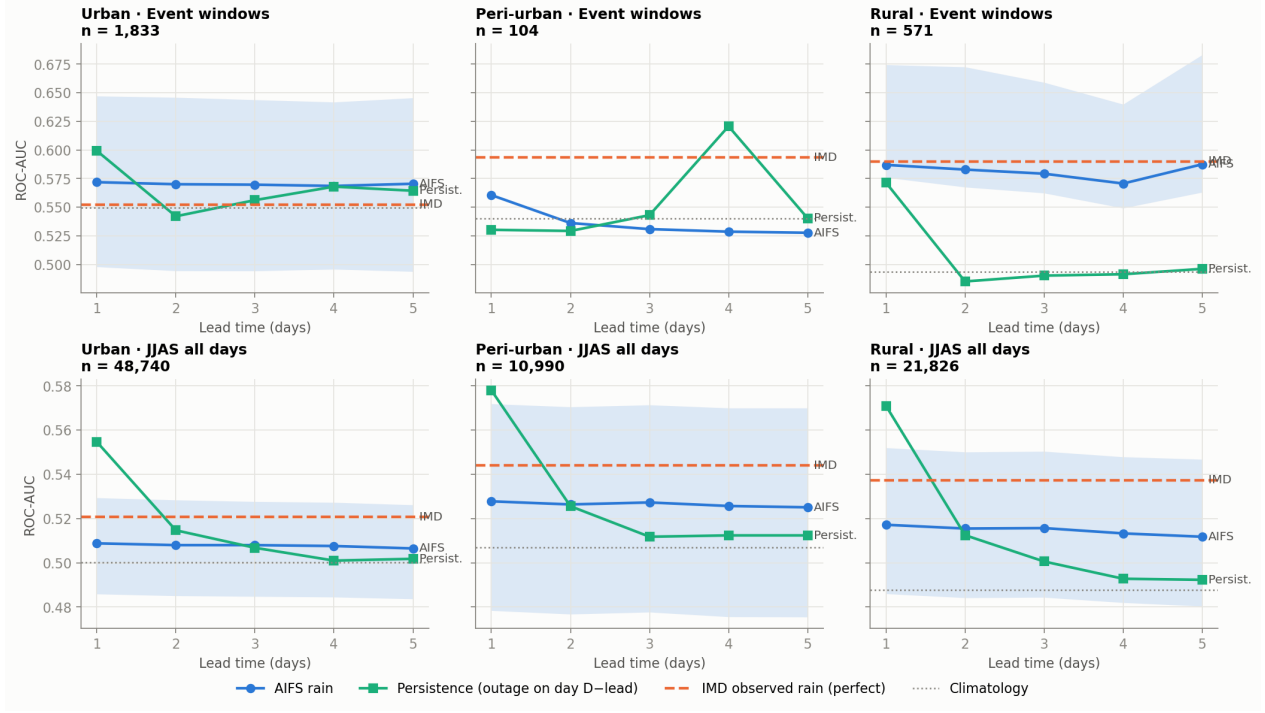


Figure 8: ROC-AUC against lead by area type. Top: event windows. Bottom: JJAS.

On flood-event windows, rural stations show the most rain skill (AIFS AUC ≈ 0.58 – 0.59 , against 0.57 for urban). Over all JJAS days, peri-urban and rural stations are similar and both slightly ahead of urban. The peri-urban event sample is tiny ($n = 104$) and all intervals overlap.

5.7 Skill by episode

Table 8: ROC-AUC by flood episode (event-only training). “–”: no disruption days, so AUC is undefined.

Ep.	GFD IDs	Country	Dates	Stn-events	n	Base %	Clim.	IMD	Pers. L1	AIFS L1	AIFS L5
1	4251	India	17 May 2015 – 21 May 2015	1	11	0.0	–	–	–	–	–
2	4282	India	15 Jul 2015 – 19 Aug 2015	2	84	1.2	0.380	0.651	0.373	0.566	0.651
3	4309	India	10 Nov 2015 – 04 Dec 2015	5	145	5.5	0.536	0.563	0.581	0.574	0.563
4	4378+4382	Bangladesh/India	07 Jul 2016 – 26 Aug 2016	26	664	12.3	0.588	0.610	0.640	0.614	0.606
5	4459	Bangladesh	30 Mar 2017 – 18 Apr 2017	9	234	9.8	0.481	0.631	0.573	0.633	0.611
6	4483	India	02 Jun 2017 – 03 Jul 2017	10	380	15.8	0.543	0.512	0.560	0.550	0.561
7	4507+4508	Bangladesh/India	10 Aug 2017 – 26 Aug 2017	18	299	17.4	0.479	0.495	0.501	0.503	0.494
8	4665	India	02 Aug 2018 – 10 Aug 2018	5	75	10.7	0.506	0.425	0.543	0.619	0.573
9	4673	India	01 Sep 2018 – 07 Sep 2018	48	616	12.0	0.483	0.533	0.577	0.516	0.502

Skill varies strongly between episodes. Episodes 1 and 2 have 0 and 1 disruption days, so their AUCs are undefined or meaningless. Among the rest, skill is highest for the 2016 monsoon floods (episode 4, AUC ≈ 0.61) and the April 2017 Assam/Bangladesh event (episode 5, ≈ 0.63). It is absent for the August 2017 floods (stations in Guwahati, Saharsa in Bihar, and Uttar Pradesh) and the September 2018 event in northern India (mostly Uttar Pradesh), which contributes 48 of the 124 station-events.

5.8 Sensitivity checks

Table 9: ROC-AUC under different spatial aggregations.

Sample	Model	Centre cell	3×3 mean	3×3 max
Event windows	Climatology	0.538	0.538	0.538
Event windows	IMD, day D	0.566	0.562	0.558
Event windows	Persistence, lead 1	0.591	0.591	0.591
Event windows	AIFS 24 h, lead 1	0.576	0.573	0.570
Event windows	AIFS 24 h, lead 3	0.569	0.569	0.568
Event windows	AIFS 24 h, lead 5	0.573	0.572	0.571
Event windows	AIFS + persistence, lead 1	0.617	0.615	0.613
JJAS all days	Climatology	0.498	0.498	0.498
JJAS all days	IMD, day D	0.528	0.528	0.525
JJAS all days	Persistence, lead 1	0.563	0.563	0.563
JJAS all days	AIFS 24 h, lead 1	0.514	0.514	0.512
JJAS all days	AIFS 24 h, lead 3	0.513	0.513	0.511
JJAS all days	AIFS 24 h, lead 5	0.511	0.511	0.509
JJAS all days	AIFS + persistence, lead 1	0.572	0.572	0.570

Table 10: Event days only (GFD start to end, no ± 3 -day padding), event-only training ($n = 1,849$).

Model	Brier	BSS	95% CI	AUC	95% CI
Climatology	0.1089	0.000	[0.000, 0.000]	0.540	[0.481, 0.617]
IMD, day D	0.1087	0.003	[-0.004, 0.007]	0.553	[0.506, 0.623]
IMD, 48 h	0.1090	-0.001	[-0.007, 0.004]	0.542	[0.491, 0.612]
IMD, lags 0–2	0.1089	0.000	[-0.005, 0.004]	0.547	[0.505, 0.617]
Persistence, lead 1	0.1077	0.011	[0.000, 0.022]	0.585	[0.542, 0.642]
Persistence, lead 2	0.1091	-0.001	[-0.007, 0.002]	0.532	[0.476, 0.605]
Persistence, lead 3	0.1091	-0.001	[-0.011, 0.005]	0.539	[0.473, 0.610]
Persistence, lead 5	0.1086	0.003	[-0.005, 0.009]	0.549	[0.490, 0.620]
AIFS 24 h, lead 1	0.1079	0.009	[-0.000, 0.019]	0.572	[0.513, 0.650]
AIFS 24 h, lead 2	0.1079	0.009	[-0.001, 0.021]	0.572	[0.512, 0.651]
AIFS 24 h, lead 3	0.1080	0.008	[0.000, 0.018]	0.569	[0.511, 0.652]
AIFS 24 h, lead 4	0.1080	0.008	[-0.001, 0.018]	0.567	[0.507, 0.648]
AIFS 24 h, lead 5	0.1079	0.010	[-0.003, 0.021]	0.571	[0.507, 0.655]
AIFS 48 h, lead 2	0.1087	0.003	[-0.003, 0.007]	0.552	[0.496, 0.631]
AIFS 48 h, lead 5	0.1085	0.004	[-0.003, 0.010]	0.557	[0.494, 0.639]
AIFS + persistence, lead 1	0.1071	0.017	[0.006, 0.032]	0.608	[0.563, 0.673]
AIFS + persistence, lead 3	0.1083	0.006	[-0.001, 0.014]	0.570	[0.509, 0.646]
AIFS + persistence, lead 5	0.1077	0.012	[-0.001, 0.023]	0.579	[0.514, 0.657]

Table 11: Event windows, models trained on all other days instead of event windows only ($n = 2,508$).

Model	Brier	BSS	95% CI	AUC	95% CI
Climatology	0.1090	0.000	[0.000, 0.000]	0.538	[0.492, 0.615]
IMD, day D	0.1082	0.007	[-0.005, 0.016]	0.568	[0.519, 0.633]
IMD, 48 h	0.1085	0.004	[-0.007, 0.013]	0.555	[0.505, 0.626]
IMD, lags 0–2	0.1082	0.007	[-0.004, 0.015]	0.574	[0.529, 0.633]
Persistence, lead 1	0.1070	0.018	[0.001, 0.033]	0.604	[0.552, 0.665]
Persistence, lead 2	0.1087	0.003	[-0.007, 0.011]	0.554	[0.491, 0.623]
Persistence, lead 3	0.1085	0.004	[-0.002, 0.009]	0.553	[0.509, 0.623]
Persistence, lead 5	0.1086	0.004	[0.001, 0.006]	0.551	[0.502, 0.625]
AIFS 24 h, lead 1	0.1083	0.006	[-0.003, 0.013]	0.565	[0.517, 0.644]
AIFS 24 h, lead 2	0.1083	0.006	[-0.003, 0.013]	0.560	[0.512, 0.641]
AIFS 24 h, lead 3	0.1083	0.006	[-0.003, 0.013]	0.559	[0.512, 0.638]
AIFS 24 h, lead 4	0.1084	0.005	[-0.003, 0.012]	0.557	[0.509, 0.636]
AIFS 24 h, lead 5	0.1083	0.006	[-0.003, 0.012]	0.557	[0.509, 0.637]
AIFS 48 h, lead 2	0.1085	0.004	[-0.004, 0.010]	0.551	[0.502, 0.630]
AIFS 48 h, lead 5	0.1085	0.004	[-0.004, 0.009]	0.548	[0.499, 0.629]
AIFS + persistence, lead 1	0.1065	0.022	[0.004, 0.041]	0.620	[0.567, 0.681]
AIFS + persistence, lead 3	0.1080	0.009	[-0.004, 0.020]	0.572	[0.528, 0.641]
AIFS + persistence, lead 5	0.1080	0.009	[-0.001, 0.016]	0.569	[0.517, 0.644]

The conclusions hold across all three checks. The spatial choice changes AUCs by less than 0.01: the centre cell is marginally highest and the 3×3 maximum marginally lowest. Training on all days helps persistence slightly and leaves AIFS unchanged.

6 Caveats and limitations

1. **ESMI data quality.** The authors of the reference study abandoned the flood–outage analysis because of data quality. Monitor failures that record 0 V are counted as outages, and monitor downtime (missing rows) could correlate with floods. The strict completeness rule removes incomplete windows, but not spurious zero readings.
2. **Weak outcome signal.** Outages are dominated by station-specific baselines and slow regime changes (e.g. episode 4). A station-level 90th-percentile threshold absorbs the level but not these changes, which dilutes the flood signal.
3. **Small number of independent events.** There are only 9 episodes, and one (Sept 2018, UP) contributes 39% of the station-events. The confidence intervals are wide and several include zero skill.
4. **Flood matching depends on choices.** The river-channel mask removes about 45% of the plain-rule station-events. The thresholds (≥ 3 events, $\geq 1 \text{ km}^2$, 20 km) are judgement calls, and Guwahati still matches three events on about 2 km^2 of residual flooding. MODIS footprints also miss short-lived urban flooding under cloud.
5. **Geocoding precision.** 40 of the 84 kept stations are placed at city centroids (errors of roughly 5–10 km). This is adequate for 20 km buffers and the 0.25° rain grid, but it blurs distance-based effects.
6. **Time alignment.** The IMD day and the ESMI day overlap by only 15.5 h. The lead-5 total is extrapolated from 21 h, and the half-step split assumes uniform rain within each 6 h step.
7. **Raw forecasts only.** AIFS is used without bias correction or ensembles. There is no IFS comparison, and 8 initialisation dates are missing (their rows are dropped).

8. **Model simplicity.** A single log-rain term with a climatology offset. Nonlinear or threshold effects (e.g. exceedance of a local rain percentile), outage counts, and interactions with grid characteristics are not explored.
9. **Outcome not validated against reports.** There is no independent record of flood-related grid shutdowns (e.g. utility press releases) against which to check the disruption days.

7 Possible next steps

1. Hand-fix the 40 kept city-centroid stations (`data/locations_manual.csv`) and re-run scripts 04–10.
2. Use an outcome less sensitive to regime changes: excess minutes over a 30-day rolling median, and a count or hurdle model for outage minutes.
3. Use extreme-rain predictors (exceedance of the station’s 95th/99th rain percentile) instead of $\log(1 + \text{rain})$.
4. Sensitivity checks: the plain 20 km rule, Domestic connections only, and 0 V-only outages.
5. Add IFS or an AIFS ensemble if 2014–2018 hindcasts become available.

A Pipeline and reproducibility

Script	Output
00_redownload_esmi.sbatch	Complete ESMI CSVs from Dataverse
01_esmi_profile_daily.py	<code>data/esmi_daily.csv.gz</code> , <code>esmi_hourly.csv.gz</code> , file profile
02_geocode_locations.py	<code>data/locations.csv</code> , review list, station map
03_subset_aifs_india.sbatch	<code>data/aifs_india/*.nc</code> (precipitation, India)
04_match_flood_events.py	<code>data/flood_events.csv</code> , <code>flood_event_stations.csv</code> , flood map
05_check_aifs_units.py	<code>logs/aifs_units_check.log</code>
06_build_panel.py	<code>data/panel_daily.parquet</code>
07_model_skill.py	<code>outputs/07_*.csv</code> ; CV predictions <code>data/cv_pred_*.parquet</code>
08_figures.py	<code>outputs/figures/08_*.png</code>
09_make_notebook.py	<code>notebooks/01_aifs_outage_skill.ipynb</code>
10_make_report.py	This report’s tables; <code>10_make_report.sbatch</code> compiles the PDF

All heavy steps ran as Slurm jobs on Sherlock (logs in `logs/`) in the owner’s `jupyter_notebook` environment. Every number in the tables is generated from the saved CSVs by `scripts/10_make_report.py`.